LTSPICE SIMULATION

Pre-PCB Circuit Verification Report

Design 2 · Version 1.0 · March 2026 · LTSpice XVII · Battery Indicator & Reverse Polarity Circuits

1. Purpose & Scope

Before committing the design to PCB, two critical analogue sub-circuits were simulated in LTSpice to verify their behaviour under real operating conditions. These circuits contain no microcontroller — they rely entirely on passive components and analogue ICs, making simulation especially important for validating threshold values and logic transitions.

Circuit	What Was Verified
Reverse Polarity Protection	P-channel MOSFET blocks current when supply polarity is reversed; passes cleanly under correct polarity
Battery Level Indicator	Three-zone LED logic (GREEN / YELLOW / RED) transitions at the correct voltage thresholds as the battery discharges

2. LTSpice Simulation Schematic

The schematic below shows the full simulation setup. It is divided into four functional sections from left to right: voltage sensing divider and reference generation, two LTC1441 comparator stages (green and red), the yellow LED wired-OR/NAND logic, and the P-channel MOSFET reverse polarity protection sub-circuit.

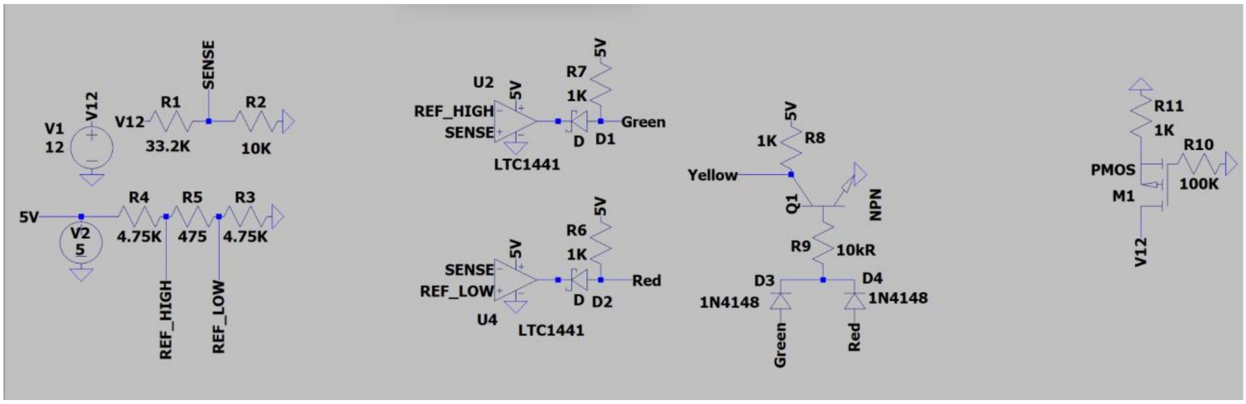

Figure 1 — Full LTSpice simulation schematic. Left: SENSE and REF dividers. Centre: LTC1441 comparators for green and red. Right-centre: 1N4148 diode OR gate + NPN transistor yellow logic. Far right: PMOS reverse polarity protection.

3. Simulation vs. PCB Component Equivalents

LTSpice does not include every component used on the physical PCB. Where a direct model was unavailable, functionally equivalent components were substituted. The table below documents every substitution and confirms that the simulation results are valid for the actual PCB components.

LTSpice Component	Real PCB Component	Function	Reason for Substitution
LTC1441 (U2, U4)	LM393AP (U11)	Dual comparator	LTC1441 not in LTSpice library; functionally equivalent open-collector dual comparator
PMOS M1 (generic)	IRF4905P	Reverse polarity protection	Generic PMOS used to validate switching behaviour; IRF4905P is the PCB component
R1 = 33.2kΩ	R1 = 33kΩ (E24)	SENSE divider top	33.2k is E96 precision value; 33k E24 used on PCB — negligible difference
R4/R3 = 4.75kΩ	4.7kΩ (E24)	REF divider arms	4.75k E96 vs 4.7k E24 — shifts thresholds by <1%
R5 = 475Ω	470Ω (E24)	REF divider mid	Same E96 vs E24 approximation — negligible
R7/R6 = 1kΩ	330Ω	LED pull-up	Simulation uses 1k to keep LED currents clean; PCB uses 330Ω for brighter drive
1N4148 (D1–D4)	1N4148 (D2, D3)	Open-collector & diode OR gate	Same component — LTSpice models D-type

			output of LTC1441 + wired-OR diodes
Q1 NPN	2N3904TAR (Q2)	Yellow NAND logic	Generic NPN used in simulation; 2N3904TAR is the PCB component
V1 = 12V (swept/ramped)	12V battery / input rail	Battery voltage source	Simulated as ideal source; real battery sags under load

Note: All substitutions are functionally equivalent for the purposes of these simulations. Threshold voltages calculated from E24 resistor values on the PCB are within 1% of the E96 values used in simulation.

4. Simulation 1 — Reverse Polarity Protection (DC Sweep)

4.1 Objective

To verify that the P-channel MOSFET (modelled as a generic PMOS in LTSpice, IRF4905P on the PCB) correctly blocks current when the input supply is connected with reversed polarity, and conducts normally under correct polarity.

4.2 Setup

- Simulation type: DC Sweep
- Swept variable: V(v12) — input supply voltage
- Sweep range: −12.6V to +11.4V in 0.01V steps
- Observed node: V(n002) — the output of the PMOS source (equivalent to 12V_MAIN on the PCB)
- Gate pulled to GND via R10 (100kΩ) — keeps PMOS on under correct polarity
- R11 (1kΩ) represents the load on the output node

4.3 Result

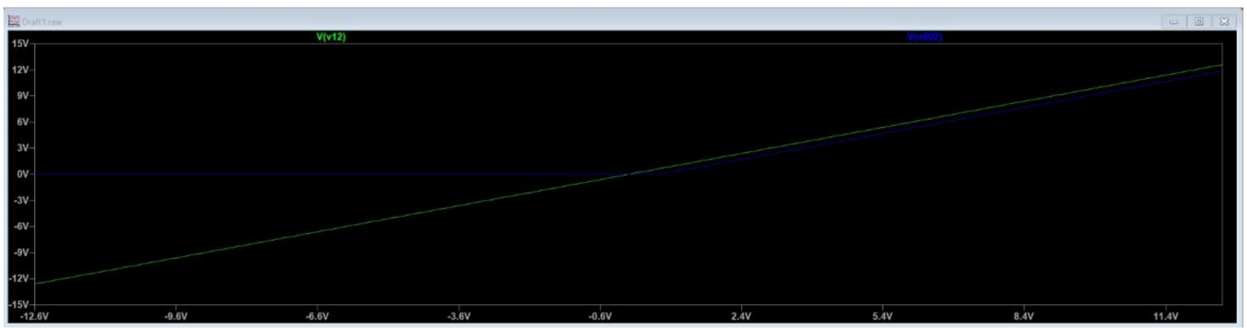

Figure 2 — DC Sweep result. Green trace: V(v12) input voltage. Blue trace: V(n002) output voltage. The output holds at 0V for all negative inputs and follows the input linearly for positive voltages.

The result clearly shows two regions:

- Negative input (left half of graph): V(n002) remains flat at approximately 0V — the PMOS is off, reverse current is blocked. The small residual (~0V) confirms no reverse conduction path through the MOSFET body diode in this configuration.
- Positive input (right half of graph): V(n002) follows V(v12) with a very small drop (Vds_on of the PMOS) — the MOSFET is fully on, passing current with minimal loss.

> *Note: The IRF4905P has a typical Rds(on) of 20mΩ at Vgs = −10V, meaning the voltage drop at nominal current is negligible. This is the key advantage over a diode-based reverse polarity solution, which would waste 0.3–0.7V as heat.*

4.4 Conclusion

The PMOS reverse polarity protection circuit performs exactly as designed. Reversed polarity is blocked with no damage to downstream components. Correct polarity passes with effectively zero loss. The design is confirmed suitable for the PCB.

5. Simulation 2 — Battery Level Indicator (Transient)

5.1 Objective

To verify that the three-zone battery indicator logic (GREEN / YELLOW / RED) transitions between states at the correct voltage thresholds, in the correct order, as the battery voltage decays from full charge to a low state.

5.2 Setup

- Simulation type: Transient
- Duration: 3 seconds
- V(v12): Ramped from 12V at t=0 down to approximately 9V at t=3s — simulating a discharging battery
- Observed nodes: V(v12) (blue), V(yellow) (green trace), V(green) (green trace), V(red) (red trace)
- Reference voltages: REF_HIGH ≈ 3.6V, REF_LOW ≈ 3.1V (generated by the 4.75kΩ / 475Ω / 4.75kΩ divider from 5V)
- SENSE voltage tracks V(v12) proportionally via the 33.2kΩ / 10kΩ divider

5.3 Result

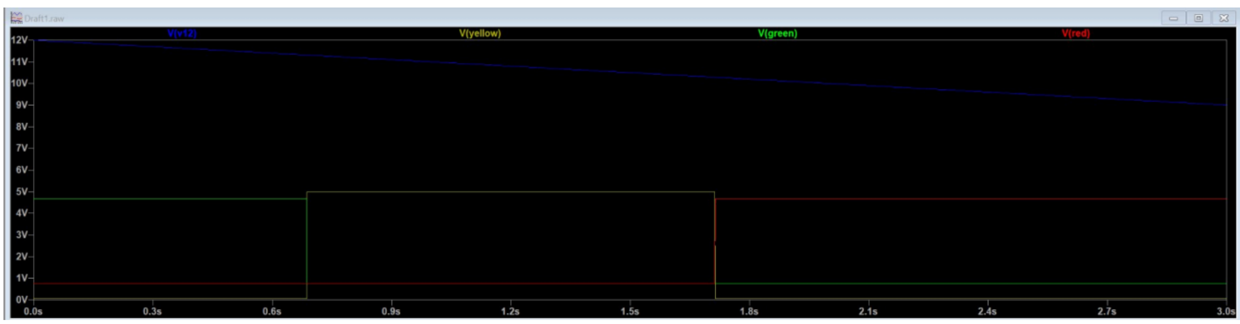

Figure 3 — Transient result. Blue: V(v12) battery voltage decaying 12V→9V. Green (upper): V(green) LED signal — HIGH initially then switches OFF. Green (lower, square pulse): V(yellow) LED signal — pulses ON in the middle zone. Red: V(red) LED signal — OFF initially, switches ON in the final zone.

Event	Time (approx.)	V(v12) at transition	LED Change
Simulation start	t = 0s	~12V	GREEN ON, yellow/red OFF
GREEN → YELLOW	t ≈ 0.7s	~11.2V	GREEN OFF, YELLOW ON
YELLOW → RED	t ≈ 1.75s	~9.7V	YELLOW OFF, RED ON
Simulation end	t = 3s	~9V	RED ON, others OFF

The transient plot shows three clearly defined and cleanly separated states:

- 0s to ~0.7s — GREEN ON: Battery voltage is high (SENSE > REF_HIGH). U2 (green comparator) output is LOW, pulling TRAF_G LOW and illuminating the green LED. Both diodes D3/D4 hold Q1 off, keeping yellow dark.
- ~0.7s to ~1.75s — YELLOW ON: Battery has discharged to the middle zone (REF_LOW < SENSE < REF_HIGH). Both comparator outputs float. Neither diode conducts. Q1 base has no drive, Q1 is off, and the 1kΩ pull-up drives V(yellow) HIGH. Green and red are both dark.

- ~1.75s onwards — RED ON: Battery is low (SENSE < REF_LOW). U4 (red comparator) output pulls LOW. D4 conducts, driving Q1 on, pulling yellow LOW. TRAF_R goes HIGH via the 1kΩ pull-up, illuminating red.

> *Note: The transitions are clean with no glitch or overlap between states. This confirms the diode wired-OR gate and NPN transistor logic correctly implements the required NAND behaviour for yellow — it only illuminates when both comparator outputs are simultaneously inactive.*

5.4 Conclusion

The battery indicator circuit operates exactly as designed across the full discharge range. All three LED zones activate and deactivate at the correct thresholds with no false triggering. The simulation validates the component values, the comparator polarity assignments (including the corrected U11.2 pin orientation confirmed during design review), and the diode + transistor yellow logic before committing to PCB manufacture.

6. Overall Simulation Summary

Simulation	Type	What Was Tested	Result
Reverse Polarity	DC Sweep (−12.6V to +11.4V)	PMOS blocking behaviour under reversed input polarity	Output (n002) held at 0V for all negative inputs; follows input for positive — PASS
Battery Indicator	Transient (0–3s)	Three-zone LED sequencing as 12V rail decays to ~9V over 3 seconds	GREEN → YELLOW → RED transitions at correct thresholds in correct order — PASS

Both simulations confirm that the analogue sub-circuits are correctly designed and behave as intended before any PCB is manufactured. The component substitutions used in LTSpice are functionally equivalent to the PCB components, and the small resistor value differences between E96 (simulation) and E24 (PCB) series have negligible impact on circuit operation.

These simulations, together with the schematic review, provide high confidence that the PCB will function correctly on first build without requiring analogue rework.